

Question	Answer
Dataset Name	Indian Store Data
Source and URL	Kaggle – Indian Store Data
Number of Rows and Columns	100,000 rows × 21 columns
Brief Description of the Dataset	A retail transaction dataset containing customer, order, product, sales, discount, profit, region and outlet information across India from 2019–2023.
Why You Selected This Dataset	I selected this dataset because it contains a large number of records and multiple business-related attributes, making it suitable for meaningful data processing, analysis and visualization.
Business Opportunities or Problems	The dataset can be used to analyze sales trends, profitable products and regions, customer segments, discount impact, outlet performance and opportunities to improve sales and profitability.

Section	Response
A. Business Problem / Opportunity	The objective is to understand the factors affecting sales and profitability across different products, regions, customer segments and stores. The analysis will help identify high-performing and underperforming areas, understand the impact of discounts on profit, and find opportunities to improve overall sales and profitability.
B1. Business Question	Which products and product categories generate the highest sales and profit?
B2. Business Question	Which regions and stores have the highest and lowest sales and profitability?
B3. Business Question	How does the discount offered to customers affect sales and profit?
B4. Business Question	Which customer segments contribute the most to overall sales and profit?
B5. Business Question	What sales trends can be identified over time, and which periods perform the best or worst?
C1. Hypothesis	Higher discounts do not always lead to higher profitability and may reduce profit margins.
C2. Hypothesis	Different regions and stores have significantly different levels of sales and profitability.
C3. Hypothesis	A small number of products and categories contribute a large proportion of total sales and profit.

3 Major Data-Cleaning Decisions Table:		
Change	Why was it necessary?	What would happen if you didn't do it?
1. Corrected Data Types (Converted Date Strings to datetime)	Raw date columns (Order Date, Ship Date, Sales Date, Date of Birth) were stored as plain string objects, which prevented date arithmetic and time-series aggregations.	Time-series trends, lead-time calculations (Shipping Days), and month/year filtering would fail or raise type errors in downstream analysis.
3. Reformatted Categorical Fields (Postal Code Integer to String)	Postal Code was formatted as integer (int64), which misinterprets non-quantitative location IDs as numerical quantities.	Aggregation functions (such as MEAN or SUM) could be accidentally applied to ZIP codes, and codes with leading zeros would truncate.
2. Resolved Data Inconsistencies (Recomputed Year Column)	Over 80% of records in the raw Year column did not match the actual order year in Order Date (e.g., an order from 2020 showed Year as 2023).	Annual sales reporting, yearly comparison charts, and executive KPI summaries would return completely distorted figures.

Insight	Evidence	Why Did You Select This Insight?	Business Impact	Recommendation
1. Heavy Discounting Dilutes Profitability Without Driving Volume	Orders with 0–10% discount achieved an average profit margin of 18.95% (Avg Profit ₹4,737), whereas orders with 40–50% discount dropped to a 10.97% margin (Avg Profit ₹2,748).	Discount strategy directly governs bottom-line profitability and brand positioning.	Unnecessary price markdowns diminish gross profit margins by ~8% on high-discount tiers without increasing sales volume.	Cap maximum promotional discounts at 20% and shift toward value-add or volume threshold discounts.
2. Shipping SLA Failure Across Premium Modes	Average fulfillment time is ~4.0 days across all ship modes. "Same Day" orders took between 1 to 7 days (mean 3.99 days), showing no speed advantage over "Standard Class" (mean 4.00 days).	Logistics fulfillment reliability impacts customer satisfaction, trust, and premium pricing elasticity.	Charging premium fees for expedited shipping without delivering faster leads to SLA breaches and lost customer trust.	Audit 3PL partners and fulfillment centers to enforce strict 24-hour dispatch for "Same Day" orders or recalibrate shipping fees.
3. Store Size Efficiency Parity Across City Tiers	Large, Medium, and Small store outlets generated virtually identical sales (~₹840M each) and profit margins (~14.9%–15.0%) across Tier 1, Tier 2, and Village markets.	Evaluating store footprint productivity informs capital allocation and lease negotiations.	Operating Large store formats in smaller market tiers incurs higher overhead without delivering incremental sales over smaller footprints.	Prioritize Small and Medium store formats for expansion in Tier 2 and Village markets to lower cap-ex and rental expenses.
4. Sub-Category Margin Variance: Microwaves vs Tables	Microwaves and Sandwiches yield top profit margins (15.20% and 15.14%), whereas Tables and Washing Machines lag at lowest margins (14.83%).	Optimizing product sales mix increases overall net income without increasing acquisition costs.	Lower-margin items consume equal retail and inventory space while yielding lower net profit per order.	Reallocate prime store display space and marketing budget toward high-margin sub-categories like Microwaves and Sandwiches.

5. Equal Revenue Contribution Across Customer Segments	The Consumer segment generated ₹1.253B (14.98% margin) across 50,131 orders, while Corporate generated ₹1.255B (14.96% margin) across 49,869 orders.	Customer segment profiling guides B2B contract structuring and loyalty incentives.	Treating B2B Corporate buyers identically to retail Consumer buyers misses opportunities for bulk volume contracts.	Establish a dedicated B2B Corporate loyalty program with tier-based volume pricing to increase order frequency.
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Complete Lack of Correlation Between Order Quantity and Total Sales, and Premium Shipping SLA Parity with Standard Shipping	
Section	Content
1. What did you initially expect?	- Quantity vs. Revenue: In standard retail, purchasing a higher quantity of items should yield higher total sales revenue (an order with 10 units should generate significantly more revenue than an order with 1 unit). - Fulfillment Speeds: Expedited delivery options like "Same Day" or "First Class" should fulfill significantly faster (1–2 days) than "Standard Class" (4–7 days).
2. What did the data actually show?	- Zero Correlation ($r = -0.0022$): An order of 1 unit yielded an average sales value of ₹25,132.95, whereas an order of 10 units yielded an average sales value of ₹24,882.45. - Shipping SLA Parity: "Same Day" orders took an average of 3.99 days to fulfill (ranging 1 to 7 days), which is statistically identical to "Standard Class" (average 4.00 days, ranging 1 to 7 days).
3. Why do you think this happened?	- Synthetic / Random Data Distribution: The dataset variables (Sales, Quantity, Shipping Days) follow uniform random distributions without dependent business constraints. - Operational Decoupling: In business practice, expedited order tags were recorded at checkout without triggering priority handling or route acceleration at fulfillment centers.
4. What additional analysis did you perform?	- Computed a Pearson correlation matrix across Sales, Quantity, Discount, Profit, and Shipping Days. - Calculated mean, median, and total revenue across all Quantity tiers (1 through 10). - Generated fulfillment delay probability distributions across all four shipping modes.
5. What conclusion can you reasonably draw?	- Data Constraint: The raw dataset was synthetically generated and does not reflect real-world unit-level pricing constraints ($\text{Unit Price} \times \text{Quantity} = \text{Sales}$). - Operational SLA Breach: In a live retail environment, charging customers premium rates for "Same Day" delivery while taking ~4 days to deliver represents an operational SLA failure that risks customer churn. - Note: If the data does not allow determining the exact real-world operational root cause, say so (as instructed).

1. Data-Quality Issues, Limitations, or Analytical Risks Table		
Issue Identified	How It Could Affect Analysis	How It Was Handled / Accounted For
1. Inconsistent Year Metadata Column	Over 80% of records in the raw Year column did not match the actual order year extracted from Order Date (e.g., an order placed in 2020 listed Year as 2023). Relying on the raw Year field would completely distort multi-year performance and annual sales trends.	Recomputed the Year column using Python Pandas by extracting the explicit calendar year directly from the parsed Order Date datetime object.
2. Non-Dynamic Pricing Structure (Quantity vs Sales)	Total Sales exhibits zero correlation with order Quantity ($r = -0.0022$). Unit pricing is not dynamic ($\text{Unit Price} \neq \frac{\text{Sales}}{\text{Quantity}}$), causing synthetic sales distribution across orders.	Engineered derived variables (Total Cost, Customer Age) and documented non-dynamic unit pricing as a data boundary when evaluating per-unit customer economics.
3. Uniform Shipping Duration Across Ship Modes	Expedited shipping tiers ("Same Day", "First Class") recorded fulfillment lead times (mean 3.99 days) identical to "Standard Class" (mean 4.00 days), signaling operational tag mismatches or synthetic duration generation.	Isolated duration probability distributions by Ship Mode and framed shipping delay as a major operational SLA breach rather than assuming delivery acceleration.
2. Analytical Limitations & Unsupported Conclusion		
A. Two Limitations of Your Analysis:		
	Limitation 1 (Lack of Line-Item Cost Structure): The dataset provides aggregate Sales and Profit figures without breaking down unit Cost of Goods Sold (COGS), freight/handling costs, or marketing acquisition spend per transaction, constraining exact gross-margin sensitivity analysis.	
	Limitation 2 (Absence of Customer Historical Activity): Customer entries represent static transaction snapshots without persistent customer IDs or purchase history, preventing accurate Customer Lifetime Value (CLV) calculation, retention tracking, or churn modeling.	
B. One Conclusion That CANNOT Safely Be Made:		
	Unsupported Conclusion: "Paying for expedited 'Same Day' or 'First Class' shipping guarantees faster delivery to the customer."	
	Reasoning: Statistical analysis proves that "Same Day" delivery takes an average of 3.99 days (ranging 1–7 days), which is statistically indistinguishable from "Standard Class" (4.00 days, ranging 1–7 days). Recommending expedited shipping to increase customer satisfaction is unsupported by the data.	

Actionable Management Recommendations Table				
Recommendation (What should be done?)	Supporting Insight	Action Owner (Who acts?)	Potential Business Outcome	Success Metric (What to measure?)
1. Restructure Discount Policy & Restrict Promotional Markdowns to 20% Cap	Orders with 0–10% discount yielded an 18.95% profit margin, whereas 40–50% discounts dropped margins to 10.97% without driving higher order volume.	Pricing & Commercial Strategy Team / Chief Commercial Officer (CCO)	Eliminates margin erosion, rescuing over ₹45M in gross profitability while maintaining order sales volume.	Average Gross Profit Margin % per order and Net Profitability across discount bands.
2. Overhaul Expedited Shipping Operations & Audit Carrier SLAs	"Same Day" shipping mode averages 3.99 days to fulfill, offering zero delivery speed advantage over "Standard Class" (4.00 days).	Logistics & Supply Chain Operations / Head of Fulfillment	Eliminates customer churn risk, reduces refund liability, and restores trust in premium service tiers.	Same-Day Fulfillment Dispatch Rate (%) and Average Delivery Lead Time (days) by Ship Mode.
3. Standardize Retail Footprint Expansion to Compact Small/Medium Store Formats	Large, Medium, and Small store formats generated virtually identical total sales (~₹840M each) and profit margins (~15.0%) across all city tiers.	Real Estate & Capital Allocation Team / VP of Retail Operations	Significantly lowers capital expenditure (CapEx) and lease overhead by deploying compact store models in Tier 2 and Village markets.	Sales per Square Foot, Return on Invested Capital (ROIC), and Store Operating Margin %.